ANN STEFFIE MADIGO

AI Leader | AI Solutions Architect & AI Engineer | Generative AI, Agentic AI & MLOps | Banking & Financial Services

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Application: Head of Artificial Intelligence | Ref. HAI/IID/2026

PROFESSIONAL SUMMARY

AI leader and hands-on AI Solutions Architect in the Co-operative Bank of Kenya's Digital Factory and Enterprise Architecture team. I authored the Bank's AI strategy and roadmap (Unified AI Integration Roadmap) and lead the architecture and delivery of its Generative AI and Agentic AI solutions, from ideation and experimentation to production deployment, across customer service, fraud management, KYC and identity risk, and digital onboarding. Experience spans banking, financial services and fintech in a highly regulated environment (CBK, Kenya Data Protection Act). I set enterprise AI architecture, engineering standards, MLOps practices and Responsible AI governance, and partner with business, risk, engineering and vendor teams to turn AI into measurable business outcomes. Building with technology since 2015; AI and machine learning projects since 2021. M.Tech in Computer Technology (in progress); AWS Certified Data Engineer and Developer. Ready to build the Bank's AI Centre of Excellence and grow its AI talent.

CORE COMPETENCIES

AI Strategy & Roadmap • Generative AI • Agentic AI • Large Language Models (LLMs) • RAG • AI Use-Case Identification & Prioritisation • Enterprise AI Architecture • AI Engineering Standards • MLOps • Production Monitoring • Reusable AI Services • Cost Optimisation • Responsible AI • AI Governance • Security, Fairness, Transparency & Accountability • Regulatory Compliance • Business Transformation • Adoption & Change Management • Data Foundations & Data Governance • Machine Learning • Computer Vision • NLP • Fraud Management • Customer Service AI • Stakeholder Management • AI Literacy & Capability Building

KEY ACHIEVEMENTS ALIGNED TO THE ROLE

- **Drive AI Strategy:** Authored the Bank's Unified AI Integration Roadmap (Nov 2025): AI as shared, reusable services across 9 bank products, organised into four capability clusters (conversational AI, document intelligence, predictive analytics, personalisation) with reference architectures and a recommended stack.
- **Accelerate AI Innovation:** Identified and delivered high-impact AI use cases in customer service (Rafiki GenAI assistant), fraud management (anomaly and structuring detection on real transactions), KYC and identity risk (face verification, OCR) and onboarding; wrote AI proposals for business owners, e.g. predictive KYC-failure risk scoring for staff onboarding.
- **Lead AI Delivery:** Led AI delivery from ideation through experimentation to production: Rafiki to pre-production for a 6M+ customer base; multi-rotation birth-certificate OCR live in production; identity verification pilot with 72.1% auto-approval at 81.2% average match confidence. Coordinated cross-functional teams across engineering, DevOps, payments, QA, vendors and business stakeholders.
- **Establish AI Standards:** Defined enterprise AI architecture patterns and engineering standards: SLA-based performance testing (12 categories, 111 automated cases), EA-approved production database sizing, CI/CD ML pipelines with experiment tracking (GitHub Actions, MLflow) and infrastructure as code (Terraform). Cost optimisation: retries run only when model confidence is low; led the evaluation of a lower-cost face-verification alternative to a commercial vendor.
- **Champion Responsible AI:** Took Rafiki through an independent security review (21 findings) and remediation; deferred high-risk agentic account opening until controls were in place; surfaced a fairness issue where a commercial KYC engine flagged older laminated National IDs as fraud; designed human-in-the-loop thresholds, audit trails and PII masking.
- **Enable Business Transformation:** Partnered with customer care, retail banking and digital teams to embed AI in processes such as balance and account-summary journeys, staff mass onboarding (CALEB) and children's account opening; used outcomes such as auto-approval rates and drop-off at the document step to steer fixes.
- **Strengthen Data Foundations:** Former data engineer (AWS Glue, Redshift, Kafka, PostgreSQL); built governed data pipelines feeding AI, including a 2.5M-record identity data pipeline and a fraud dataset with PII masked before analysis.
- **Build AI Capabilities:** Champion AI literacy: plain-language AI explainers for non-technical stakeholders, developer handover packs and design standards; teaching assistant for AI units at the Technical University of Kenya (2017–2022) and at eMobilis; co-founded a school computer club teaching mobile programming (2015–2016).

PROFESSIONAL EXPERIENCE

Solutions Architect / AI Engineer — AI Enablement (Contract)

Co-operative Bank of Kenya, Digital Factory & Enterprise Architecture, ICT & Innovation Division, Nairobi | Oct 2025 – Present

AI strategy and governance

- Authored the Unified AI Integration Roadmap: shared platform services (LLM gateway, RAG knowledge bases, document intelligence, scoring APIs) with common guardrails, logging and audit, covering YEA, Co-op Bank App, WhatsApp banking, CALEB, Co-op Soko, Bancassurance, Asset Finance and fraud.

Rafiki — Generative AI and Agentic AI banking assistant (customer service)

- Architected and built Rafiki on AWS Bedrock (Claude) with a Bedrock Knowledge Base (RAG on OpenSearch Serverless), Kafka event streaming, Next.js, Docker and Kubernetes/ArgoCD; deployed to pre-production for the YEA/IGNITE youth segment (6M+ customers).
- Multilingual NLP: automatic language detection and two-pass English/Swahili/Sheng translation; WhatsApp Cloud API with HMAC-verified webhooks; M-Pesa STK push.
- Designed the Rafiki AI Channel: a dedicated AI Engine and WhatsApp Gateway with National ID + Chatbot PIN + OTP authentication; passive liveness detection with active liveness for escalations.

Identity verification and KYC (computer vision)

- Built the identity verification platform: async Python pipeline over ~2.5M records; Amazon Textract OCR, face matching (Rekognition with an ArcFace/SFace ensemble), tamper detection and fraud scoring on AWS Lambda and Step Functions.
- Led the vendor evaluation (AWS vs GBG IDscan) and the build-vs-buy recommendation: neither vendor was production-ready end to end, so recommended an AWS-based platform now covering all Kenyan ID types.

Birth-certificate OCR for children's accounts (Jumbo Junior)

- Shipped multi-rotation handling in the Bedrock OCR handler (Claude vision), live in production: when the extracted name looks implausible and confidence is below 70%, the certificate is re-read in three orientations, so sideways photos no longer block parents.
- Diagnosed production timeouts and an image-orientation bug with engineering; fixed extraction that returned the registrar's name instead of the child's; investigated a fraud case (23+ accounts from one edited certificate) and prototyped tamper checks.

Fraud management (machine learning)

- Built streaming fraud detection (Django, Kafka, K-means anomaly scoring, analyst review queue; sub-100ms latency target), extended to approved real transaction data to detect structuring/cycling patterns.

Solution architecture

- Led architecture for CALEB (staff mass account opening): sized production for 2,100 accounts/day and ~4,000 concurrent branch staff; defined the 111-case performance test suite. Authored 17+ solution designs across payments, authentication and onboarding.

AWS Developer & Data Engineer

A1 Strategic Business Solutions, Nairobi | *Aug 2024 – Sep 2025*

- Built AWS data pipelines (Glue, Redshift, QuickSight), cutting monthly reporting time by ~60%.
- Built Generative AI invoice automation (Amazon Textract + Bedrock LLMs, Step Functions) to extract, classify and validate invoices.
- Prototyped machine learning models for credit scoring and fraud detection (Amazon SageMaker).
- Designed a Bedrock RAG conversational-AI architecture for a 7-channel banking tender.

Data Engineer Intern (Fintech)

ViewTech Limited / SasaPay, Nairobi | *Jun 2023 – May 2024*

- Built mobile-money transaction data pipelines and PostgreSQL schemas for transactions, customers and reporting.

AI Engineer Intern

Miradi Robotics, Nairobi | *2020 – 2022*

- Worked on AI and blockchain features for a home raffle management system.

SELECTED AI & MACHINE LEARNING PROJECTS

MaterialsGPT Africa — materials design copilot (2026, in progress) | *LLM agents, conditional VAE, graph neural network*

- Lead the LLM copilot for a TU Kenya research team: turns a goal (e.g. a solar water-treatment material) into target properties, calls generative and GNN models, ranks candidates and explains each with cited evidence.

Wellness Tourism Purchase Prediction — end-to-end MLOps (2026) | *XGBoost, MLflow, GitHub Actions, Streamlit*

- Automated retraining pipeline with experiment tracking and a live app; ROC-AUC 0.92, recall 0.79; call conversion 59% vs 19% baseline.
  github.com/Steffie254/wellness-tourism-purchase-prediction-mlops

NewsFindr — agentic AI news briefings (2026) | *LangGraph, LangChain, Groq, SQL agent*

- Agent reads user interests from SQL, searches, filters sources for credibility and writes cited briefings, with a token-budget limiter and deterministic fallback.

Healthcare RAG Assistant (2025) | *Mistral-7B, llama.cpp, LangChain, Chroma*

- Privacy-preserving, offline RAG over the Merck Manual for grounded clinical answers.
  github.com/Steffie254/healthcare-NLP-RAG-project

ReneWind — predictive maintenance with deep learning (2025) | *TensorFlow/Keras*

- 7 neural-network designs compared; best model: 98.9% accuracy, 0.87 recall, 0.93 precision on turbine failures.
  github.com/Steffie254/renewind-predictive-maintenance-dnn

Other AI projects (2021–2026) | *Keras, NLTK, Flask, SpeechRecognition, OpenCV, scikit-learn*

- AI banking FAQ chatbot with a neural intent classifier (2021); VOCO voice assistant with speech recognition and intent detection (2022); handwritten digit recognition (2022); unsupervised payment-fraud detection with clustering (2024); Airbnb pricing model (2025); Nyuki mobile AI writing app on Qwen 2.5 (2026).
  github.com/Steffie254

EDUCATION

Master of Technology in Computer Technology (in progress) | Technical University of Kenya

Research focus: AI-based biometric verification and KYC fraud in digital banking onboarding. Coursework: Advanced Algorithms, Research Methods, ICT Project Management, Information Security.

Bachelor of Technology in Computer Science, Second Class Honours (Upper Division) | Technical University of Kenya | 2017-2022

CERTIFICATIONS

- AWS Certified Data Engineer – Associate (2025)
- AWS Certified Developer – Associate (2024)
- Kaggle: Intro to Machine Learning, Intermediate Machine Learning, Intro to Deep Learning (2024)
- Meta Back-End Developer Professional Certificate; eMobilis Mobile Entrepreneurship Program
- In progress: AWS Solutions Architect – Associate

LEADERSHIP, TEACHING & COMMUNITY

- Teaching Assistant, AI-related units, Technical University of Kenya (2017-2022)
- Teaching Assistant, Mobile Entrepreneurship Program, eMobilis Institute of Technology (2017)
- Co-founder, Computer Club, Kisumu Girls High School, sponsored by eMobilis: basic mobile programming in Java (2015-2016)

TECHNICAL SKILLS

- **Generative & Agentic AI:** AWS Bedrock (Claude), RAG, Knowledge Bases, LangChain, LangGraph, open-source LLMs (Mistral, Qwen), prompt engineering, guardrails, evaluation
- **Machine Learning & Computer Vision:** scikit-learn, XGBoost, TensorFlow/Keras, anomaly detection, OCR (Textract, Claude vision), face matching (ArcFace, Rekognition), OpenCV, NLP
- **MLOps & Cloud:** MLflow, GitHub Actions, Docker, Kubernetes, ArgoCD, Terraform, AWS (Lambda, Step Functions, S3, SageMaker, Glue, Redshift)
- **Data & Integration:** Kafka, PostgreSQL, Oracle, Redis, REST/OpenAPI, microservices, event-driven architecture
- **Engineering:** Python, Django, TypeScript, Next.js, NestJS, Vue, Java
- **Governance & Compliance:** CBK guidelines, Kenya Data Protection Act, PCI-DSS, OAuth2, audit trails, human-in-the-loop controls